# Ashneet Kaur Bhatia

ashneet0402@gmail.com | +91-8827992883 | Hyderabad, India
LinkedIn

## Profile

Chemistry postgraduate currently pursuing M.Sc. Chemistry at the University of Hyderabad, with a B.Sc. Honours in Chemistry and hands on laboratory experience in analytical chemistry, sample preparation, UV Vis spectroscopy, wet chemical analysis, quantitative measurements, and laboratory documentation. Research exposure at CSIR CIMFR with emphasis on SOP adherence, measurement traceability, calibration verification, reproducibility, laboratory safety, and accurate experimental reporting. Interested in analytical chemistry, chemical characterization, water chemistry, nuclear materials chemistry, radiation chemistry, and scientific instrumentation.

## Education

**University of Hyderabad** — *2026 – Present*
*M.Sc. Honours Chemistry* — *Currently Pursuing*

**MANIT Bhopal** — *2026*
*Former Student - M.Sc. Chemistry*

**Guru Ghasidas Vishwavidyalaya, Bilaspur** — *2023 – 2026*
*B.Sc. Honours Chemistry* — *CGPA: 8.2 / 10.0*

**Khalsa Public School, Dongargarh** — *2022*
*Class XII, CBSE / State Board* — *93.4%*

**Khalsa Public School, Dongargarh** — *2020*
*Class X, CBSE / State Board* — *90.4%*

## Technical Skills

**Analytical Chemistry:** Acid base titration, redox titration, conductometric titration, gravimetric analysis, UV Vis spectroscopy, pH analysis, basic chromatography, quantitative and qualitative analysis
**Sample Preparation and Measurement:** Sample preparation, reagent preparation, standard solution preparation, dilution protocols, blank correction, calibration curves, quantitative estimation, repeat measurements
**Instrumentation:** UV Vis spectrophotometer, analytical balance, pH meter, centrifuge, muffle furnace, basic chromatographic systems
**Laboratory Practices:** SOP adherence, GLP, laboratory documentation, raw data traceability, sample identification, logbook maintenance, calibration verification, result reproducibility
**Safety and Compliance:** HSE practices, chemical handling, laboratory waste disposal, ISO/IEC 17025 awareness, ISO 9001 awareness
**Laboratory Data Systems:** LIMS basics, structured analytical records, result documentation

## Research and Laboratory Experience

**CSIR CIMFR, Bilaspur** — *May 2025 – Jun 2025*
*Research Intern* — *Bilaspur, India*

- Prepared and processed laboratory samples for routine chemical analysis using documented SOPs, maintaining consistency in sample handling and preparation.
- Maintained raw data sheets, sample labels, laboratory logbooks, and analytical records to support result traceability, reproducibility, and accurate reporting.
- Assisted in instrument calibration verification and followed laboratory safety, chemical handling, waste disposal, and documented HSE procedures.

**Department of Chemistry Laboratory, Guru Ghasidas Vishwavidyalaya** — *Aug 2023 – May 2026*
*Undergraduate Laboratory Trainee* — *Bilaspur, India*

- Performed quantitative and qualitative chemical experiments with emphasis on accurate measurements, solution standardization, repeatability, and systematic observation.

## Academic Projects

**Development and Validation of a Trace Metal Analysis Method** — *2026*

- Developed a quantitative analytical method for trace metal determination in aqueous samples using instrumental spectroscopy and standardized sample preparation procedures.
- Evaluated calibration linearity, precision, recovery, blank correction, repeatability, detection limits, and measurement uncertainty to assess analytical method performance.
- Applied quantitative data analysis and systematic documentation to improve reliability and traceability of analytical results.

**Sorption and Separation of Metal Ions from Aqueous Systems** — *2025*

- Investigated pH, contact time, concentration, and temperature dependent sorption of metal ions using a functionalized solid sorbent.
- Evaluated adsorption kinetics, equilibrium behaviour, selectivity, adsorption capacity, and regeneration using quantitative analytical measurements.
- Characterized the sorbent using available spectroscopic and material characterization techniques and interpreted experimental results.

**Water Chemistry and Corrosion Behaviour of Reactor Materials** — *2025*

- Investigated the influence of pH, chloride concentration, temperature, and dissolved species on corrosion behaviour of carbon steel or stainless steel under controlled aqueous conditions.
- Quantified corrosion behaviour using mass loss and/or electrochemical measurements and monitored changes in solution chemistry.

## Achievements

**Competitive Examinations**
- IIT JAM 2026: AIR 1708
- CUET PG 2026: AIR 9712

**Academic Achievements**
- College Science Fair: 1/60
- Inter School Debate Champion

**Leadership**
- Overall Head, GGV Science Club
- Academic Event Coordination

## Certifications

- **Advanced Transition Metal Chemistry** | IIT Bombay / NPTEL | Elite, 61%
- **Introduction to Molecular Spectroscopy** | University of Manchester / Coursera
- **Dr. K's Basics of Nuclear Chemistry** | Udemy